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MILITARY SPECIFICATION

GRINDING MACHINES, INTERNAL, CHUCKING, HORIZONTAL, ELECTRIC MOTOR DRIVEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric motor driven, floor mounted, horizontal spindle, internal, cylindrical grinding machines with equipment and accessories as described herein.

1.2 Classification. This specification covers grinding machines of the following types, classes and sizes. The type, class, and size furnished shall be as specified (see 6.2.1).

Type I - Manual Sizing with Manual/Power Operation

Size	Swing Inside Standard Guard (Inches)	Grinding Stroke Range (Inches)
1	8	1/10 to 3
2	12	3/8 to 8
3	24	3/8 to 10
4	36	3/8 to 12
5	42	3/8 to 13

Type II - Semi-Automatic Sizing

Class A - Manual/Power Operation

Size	Swing Inside Standard Guard (Inches)	Grinding Stroke Range (Inches)
1	8	1/10 to 3
2	12	3/8 to 5
3	24	5/8 to 15
4	36	3/8 to 24
5	42	3/8 to 13

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3415

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Class B - Computer Numerically Controlled (CNC) Operation

Size	Swing Inside Standard Guard (Inches)	Grinding Stroke Range (Inches)
1	6	1/16 to 9
2	12	1/16 to 9
3	18	1/16 to 17
4	24	1/16 to 16
5	40	1 to 20

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless othewise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the data of the solicitation.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition

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functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of documents which are DoD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the nongoverment documents which is current on the data of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA MG 1 - Motors and Generators.

ANSI/NFPA 79 - Metalworking Machine Tools and Plastics Machinery, Electrical Standards for.

ANSI B5.8 - Chucks and Chuck Jaws.

ANSI B7.1 - Abrasive Wheels, Use, Care, and Protection of, Safety Requirements for the.

ANSI B11.9 - Safety Requirements for the Construction, Care and Use of Grinding Machines.

ANSI B74.13 - Marking for Identifying Grinding Wheels and Other Bonded Abrasives.

ANSI B74.2 - Shapes and Sizes of Grinding Wheels, and Shapes, Sizes, and Identification of Mounted Wheels, Specifications for.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D-3951 - Commercial Packaging, Standard Practice for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods for Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturer's Association, Standards Department, Suite 1000, 1901 North Fort Myer Drive, Arlington, VA 22209-1695.)

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INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO-54 - Cylindrical Gears for General Engineering and for Heavy Engineering, Modules and Diametral Pitches of.

ISO-R 1122 - Gears, Glossary of, Geometrical Definitions.

(Application for copies should be addressed to the International Organization for Standardization, c/o the American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA) STANDARDS

RS-232 - Interface between Data Terminal Equipment and Data Communication Equipment Employing Serial Binary Data Interchange.

RS-267 - Axis and Motion Nomenclature for Numerically Controlled Machine Tools.

RS-281 - Construction Standards - Numerical Machine Tool Control.

(Application for copies should be addressed to Electronic Industries Association, Engineering Department, 2001 Eye Street, N.W., Washington, DC 20006.)

(Nongovernment standards and other pulications are normally available from the organizations which prepare of which distribute the the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.2.2 and 6.3).

3.2 Design. The machine shall be new and one of the manufacturer's current models. Each machine shall be capable of precision internal cylindrical and tapered grinding to tolerances specified herein. Each machine shall incorporate in its design means to permit relative movement of workhead and wheelhead longitudinally, transversely and angularly. Such means for each movement described shall comprise movement capability of workhead, wheelhead, or both. Type I machines shall be designed with a system to manually size workpieces to predetermined dimensions. Type II machines shall be designed with a system to automatically size workpieces to set dimensions. Class A machines shall position the wheelhead table by power; class B machines shall position the wheelhead by numerical control. The machine design and construction shall include all components, parts, and features

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necessary to meet the performance requiremetns specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repairs.

3.2.1 Measurement systems. Unless otherwise specified (see 6.2.1), either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified (see 6.2.1), either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use or performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The machine and all its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the Industry.

3.2.4 Operating controls. All operating controls necessary for operation of the machine shall be mounted on either a control console or pendant stand clearly identified to facilitate its use. Voltage and current meters, indicator lights, and all other operational control aids shall be arranged and mounted convenient to and in full view of the operator. An emergency stop switch shall be furnished on the control station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance

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and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with ANSI B11.9 and with all of the requirements of OSHA 2206 that are applicable to the machine itself. In the event of a conflict between the requirements of the ANSI and OSHA standards, the OSHA standards shall take precedence. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. The machine shall be designed and constructed such that, under the operating, service, transportation and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state or local statutes in effect at the point of the installation (see 6.4).

3.2.9 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerance.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No processes such as welding, peening, plugging, or filling with solders or pastes shall be used for relaiming any defective part.

3.3.2 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to

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perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. All surfaces to be painted shall be cleaned of all foreign matter. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with manufacturer's standard commercial practice which shall be not less than one coat of primer and one coat of finish color.

3.3.6 Threads. Unless otherwise specified (see 6.2.1), all threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All rotating dials or handwheels which are used to position machine axes or for tool movement shall be graduated in increments not larger than 0.001 inch. Dial and handwheel diameters shall be permanently and legibly engraved or etched on a nonglare background with graduation that can be read from the operator's normal position.

3.3.8 Gears. All gears used in the machine or its components shall be properly selected for the intended application and fabricated to either English (US) standards per AGMA 390.03, and AGMA 360.02 or Metric (SI) standards per the International Organization for Standardization ISO-R1122 and ISO-54.

3.4 Components. The machine shall consist basically of the following components. Any additional components necessary for the operation of the machine shall be furnished in accordance with manufacturer's standard practice.

3.4.1 Bed. The bed shall be of epoxy granite, iron or steel, ribbed and braced to withstand forces imposed when the machine is utilized to its maximum capacities. Guideways shall be provided for supporting the workhead and wheelhead assemblies. When applicable, hydraulic and coolant reservoirs may be housed in the base or provided as separate external units. When applicable, an access door(s) or removable plate(s) shall be provided for servicing components housed within the base. The bed shall have not less than 3-point floor bearing. When an extended bridge type machine is required, the bed shall incorporate a pedestal mounted bridge of specified dimensions (see 6.2.1), with workhead cross slide.

3.4.2 Workhead cross slide (U axis). The workhead cross slide, when furnished, shall be constructed to withstand any work load encountered during normal operation. The slide shall have mating guideway surfaces corresponding to those of the bed. The slide shall serve as a mount for the workhead and its components and shall include means for workhead swivel to facilitate taper grinding. A graduated scale with increments not larger than one degree, with index marker and means for relocating the workhead

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from an angular position to straight position, shall be incorporated in the design of the cross slide. Cross slide travel shall be accomplished manually. When provided, the cross slide shall be either power operated or numerically controlled, as specified (see 6.2.1). A means for securing the workhead cross slide after positioning shall be provided.

3.4.2.1 Workhead cross slide feed. When furnished, workhead cross slide feed motion shall be accomplished through a precision bearing system. These bearings shall be protected in all positions of the slide to prevent the entrance of foreign matter. For manual operation, the cross slide shall be equipped with a handwheel and leadscrew having provisions for fine feed. Power or numerically controlled feed, when applicable, shall be adjustable to permit selection of the proper stock removal rate for the part to be ground. Provision shall be made to assure constant cross slide feed rate regardless of temperature fluctuations in the machine.

3.4.3 Workhead. The workhead housing shall be oil tight, and shall protect the workhead spindle and its components. The workhead shall be shielded to protect the spindle bearings from foreign matter. Means for quick shockless braking of the spindle and a device allowing the spindle to be revolved slowly when indicating or checking a workpiece shall be incorporated in the design of the workhead. When specified (see 6.2.1), the chuck operating mechanism shall be power operated. A fixed workhead, if used, shall be integral with or mounted on the base. A swiveling workhead, if used to facilitate taper correction, shall be mounted on a cross slide conforming to 3.4.2 or a fixed bridge. The workhead or base shall have a graduated scale for setting angles and a positive locking mechanism. When specified (see 6.2.1), type II, class B machines shall provide programmable automatic swivel angle positioning of the workhead.

3.4.3.1 Workhead spindle. The workhead spindle shall be of alloy steel arranged with means for accepting a chuck or other work-holding fixture at one end and a driving mechanism at the other. The spindle shall be mounted on self-adjusting bearings which are pressure lubricated with filtered oil, or in precision pre-loaded ball or roller bearings greased for life. The internal and external surfaces of the spindle nose or flange, on which the chuck or other work-holding fixtures rest, shall have a hardness not less than Rockwell C-40. Unless otherwise specified (see 6.2.1), the workhead spindle flange shall be manufacturer's standard.

3.4.3.2 Workhead spindle drive. The workhead spindle drive shall include a drive motor with belt and pulley arrangement, a spindle speed adjustment system which shall be infinitely variable throughout its entire speed range, and, when specified (see 6.2.1), a speed indicating device calibrated in RPM. Type II, class B machines shall have programmable spindle RPM and rotational direction from sequence to sequence through manual data input (MDI) and/or automatic programming routines. The drive motor shall be reversible. When a belt driven workhead is provided, means for adjusting belt tension shall be provided unless a non-slipping, non-tensioning belt is furnished.

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3.4.4 Wheel truing device. Unless otherwise specified (see 6.2.1), a grinding wheel dresser, complete with one industrial diamond, shall be rigidly mounted on the machine. Type I machines shall be provided with a manually-operated wheel dresser When specified (see 6.2.1), the truing device shall be designed to automatically lift or raise to the "rest" position when the wheelside traverses forward beyond the wheel dressing position. Type II machines shall be furnished with a wheel dresser automatically operated and compatible with the automatic sizing system. Type II, class B machines shall have programmable wheel truing cycles including dressing at cycle start, a single interruption at rough grinding cycle completion, or both; dress on an alternating cycle basis; multiple dress; or in-cycle dress by MDI (pushbutton). Appropriate dress rate shall be established by the control based on desired wheel speed and dress lead. Dress modes shall be selective from sequence to sequence. The wheel dresser shall be manufacturer's standard, readily adjustable for various work sizes and shall include means for fine adjustment either on the dresser unit or, on type II, class B machines, by control input MDI.

3.4.5 Wheelhead. The wheelhead shall be mounted to the X axis wheelhead cross slide or wheelhead plate, as applicable, and shall be either belt driven with an external motor and pulley arrangement, or the self-contained type. The wheelhead(s) shall be of the type, style and size specified, and the sizes and styles of quills or sleeves to fit the wheelhead shall be as specified (see 6.2.1). When a belt driven wheelhead is required, means for adjusting belt tension shall be provided. A wheelhead mounting plate shall allow for pivoting the wheelhead a minimum of 5 degrees.

3.4.6 Wheelhead cross slide table. The wheelhead cross slide table shall be a one-piece casting designed to support any maximum rated machine work load. The table shall be mounted on guideways provided on the bed or cross slide and arranged to traverse longitudinally (Z axis) along the bed. Means shall be provided for table movement for set-up and manual operation. The length of table stroke shall be infinitely variable within the total grinding stroke of the machine. Type I and type II, class A machines shall be provided with an adjustable mechanism for automatic traverse during the grinding stroke. Table reversal shall be automatic. Type II, class B machines shall be provided with numerical control of wheelhead table travel and reversal. The machine shall have means for preventing over-travel of the table. The wheelhead table shall include means for mounting a wheelhead and its driving components. When table design includes an adjustable, top-mounted cross slide, it shall meet the requirement in 3.4.7.

3.4.7 Wheelhead table cross slide. The wheelhead table cross slide (X axis), when furnished, shall be constructed to support the wheelhead table and its driving components. The cross slide shall have mating guideway surfaces corresponding to those of the table or bed arranged to traverse perpendicular to the table travel (Z axis), forming a compound slide arrangement. On type I and type II, class A machines, cross slide operation shall be manual or powered or both. Type II, class B machines shall have numeri-

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cally controlled wheelhead table cross slide positioning. A safety device shall be provided to prevent overtravel of the slide.

3.4.8 Automatic wheel change. When specified (see 6.2.1), type II, class B machines shall be capable of automatic wheel changing. Wheel change shall be actuated by command from the numerical control. Wheel change shall be accomplished by means of a power-operated mechanism which shall either interchange the wheel in the spindle with a wheel selected from the magazine or remove the wheel from the spindle and then insert the wheel from the magazine. Wheel change shall be possible any time during the operating cycle. The magazine shall permit convenient loading and unloading of wheels. The wheel changer shall be arranged for random wheel selection which shall include the capability of reusing a wheel that has been returned to the magazine after a program operation.

3.4.9 Grinding wheels. Grinding wheels shall conform to ANSI B74.2. The wheels shall be marked in accordance with ANSI B74.13 and shall meet the safety requirements established in ANSI B7.1.

3.4.10 Lubrication. A lubrication system shall be provided which will insure adequate lubrication for all moving parts. The system shall deliver lubrication to all ball screws, mating gears, sliding parts and bearings (except the sealed-for-life type). Each lubricant reservoir shall have a means for determining and maintaining fluid level. The lubricant reservoir shall have sufficient capability to operate for a minimum of 80 hours of feed time without being serviced. The lubrication system shall have a device for indicating oil level and, on type II, class B machines, a low level pressure warning to alert the operator to stop spindle and axis motion. Recirculated oil shall be filtered by a cleanable or replaceable type filter. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing.

3.4.11 Coolant system. Unless otherwise specified (see 6.2.1), a coolant system shall be provided. The coolant system shall include a sump, or reservoir, a power driven pump, and piping. The coolant reservoir shall be of sufficient capacity to permit full flow cooling. The coolant system shall include a baffle(s) or strainer which is easily removable for cleaning. The reservoir shall be equipped with means for emptying and cleaning. A coolant delivery nozzle and valve shall be provided to permit the operator to direct and control the amount of coolant flow. The coolant system on type II, class B machines shall be switched on and off by the numerical control system.

3.4.12 Hydraulic system. When specified (see 6.2.1), each machine shall have a hydraulic system. The hydraulic system shall be complete with pumps, valves, piping, cylinders, a reservoir, and pressure controls. Overpressure protection shall be provided on the high pressure line from the hydraulic pump to prevent damage to components. The system shall include filters to insure delivery of clean fluid. The reservoir shall have a sight gage to indicate the fluid level and a low level alert to warn the operator to stop

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spindle and axis motion. The fluid temperature shall be maintained within the proper operating range during all normal operations of the machine.

3.4.13 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 79. Each machine shall draw all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 79 for the disconnect fuse rating.

3.4.13.1 Motors. Separate motors shall drive each of the two axes, grinding wheel, work spindle, coolant pump, coolant filter and hydraulic pump, as applicable. Motors shall be single or multiple speed AC, or variable speed DC, having ball or roller bearings and rated for continuous duty. Each motor used shall be suitable for its application and, unless otherwise specified (see 6.2.1), shall meet the requirements of a dripproof enclosure in accordance with ANSI Standard NEMA MG-1.

3.4.14 Machine ways. Machine guideways shall be adequately protected wherever possible to exclude chips, dirt, coolant, and other hazards that may cause damage or result in inaccurate functioning of any members. The machine ways shall be hardened, ground, and fitted to the sliding member by a gib arrangement, or other suitable devices, to achieve the accuracies specified in table II. Ways shall be designed for use with anti-friction way materials or hydrostatic bearings.

3.4.15 Automatic sizing system. Each type II machine shall be provided with an automatic sizing system to permit automatic grinding of workpieces to predetermined dimensions. Each system shall allow for roughing and finishing of bores and dressing of wheels, all automatically. Size control shall include one or a combination of the following as specified (see 6.2.1):

a. Diamond sizing.
b. In-process plug gaging.
c. In-process air gaging.
d. In-process electronic gaging.

3.4.16 Numerical control. Type II, style B machines shall be completely programmable through a fully automatic solid state, soft-wired, integrated circuit type numerical control. The machine control unit (MCU) shall provide automatic control of machine functions, operating modes, two axis (X and Z) positioning and feed motion, workhead speed and direction of rotation for each grind surface, rapid traverse rates, axis dwell times, dress type, dress rates, dress frequency, and shall initiate action should a machine fault occur. The two axes (X and Z) shall be identified in accordance with EIA Standard RS-267. The control shall have linear interpolation and be

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capable of two axis simultaneous control. Control features shall include fixed cycles, part program storage, part program and buffer edit capability, and control diagnostics. The MCU shall accept part program data entered from a keyboard (manual data input-MDI) and from an external input device compatible with the part program storage media (cassette tape cartridge, magnetic card, etc.) of the type specified (see 6.2.1). The control unit shall conform to EIA Standard RS-281. A line voltage of ±10% from normal shall not adversely affect the numerical control function. The control shall be capable of functioning in ambient temperatures ranging from 50°F to 120°F and relative humidities ranging from 5% to 95% non-condensing. The control shall automatically shut down when internal operating temperature exceeds safe operating temperature. Control resolution shall be 0.000025 inch for each linear axis.

3.4.16.1 Data display. A cathode rate tube (CRT) or multi-line alpha-numeric display, capable of displaying at least 250 characters simultaneously, shall be provided. Unless otherwise specified (see 6.2.1), the display shall, as a minimum, display the following data:

a. Block sequence number or cycle information.
b. Actual and command position of each axis.
c. Feedrate and/or override condition.
d. Control and system error messages.

3.4.16.2 Control system modes of operation. The operational modes of the NC control shall include manual, automatic, single block, and MDI.

3.4.16.3 Switchable inch/metric input. The control system shall provide switchable inch/metric input capability.

3.4.16.4 Part program memory. Unless otherwise specified (see 6.2.1), the part program shall reside in either solid state or bubble type memories. Unless otherwise specified (see 6.2.1), the MCU shall be capable of storing in memory the content of not less than 250 instructions or 10 grind surfaces for part program data.

3.4.16.5 Standby battery power. Unless otherwise specified (see 6.2.1), standby battery power shall be provided for controls having volatile memory. The standby battery power shall be capable of retaining a fully loaded memory for a period of not less than 72 hours after the primary power source fails.

3.4.16.6 Part program edit. A program edit capability shall be provided to allow block insertion, additions, deletion or modification of a part program in the part program memory. Data related to part program edit shall be input by the manual data input provided on the control panel. Edited

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program data entered into program memory shall be automatically acted upon, in the sequence designated, whenever the part program is cycled.

3.4.16.7 Feed rate override. A feed rate override control shall be provided to allow manual adjustment of the programmed feed rate from 10 to not less than 120 percent of the programmed rate in either continuous adjustment or in steps of not more than 10 percent. The feed rate override control shall have a position that allows return of feed rate to that originally programmed.

3.4.16.8 Sequence number search. The control shall have the capability to permit the operator to search for any desired sequence number.

3.4.16.9 Block delete. The control shall have a delete feature to permit by-passing one or more instructions of programmed data.

3.4.16.10 Segmental read. The control shall have a function to provide for reading information one segment at a time.

3.4.16.11 Automatic cycle. When specified (see 6.2.1), the control shall have the capability of running in automatic cycle. Once programmed, the control shall automatically run for an infinite number of parts.

3.4.16.12 Reversal error compensation. Error resulting from reversing the direction of the leadscrew shall be minimized by the use of appropriate mechanical devices designed to eliminate as much reversal error as possible. Electronic compensation within the control unit may be used to reduce the effects of reversal error. Means shall be provided to adjust the antibacklash device and to input new compensation valves, if applicable, to correct for wear and usage during the life of the machine. The backlash eliminator/compensation system shall provide sufficient accuracy to meet the tolerances given in table II.

3.4.16.13 Axis calibration. Error compensation shall be provided, as necessary, to correct repeatable errors determined to exist due to leadscrew and feedback inaccuracies. A means shall be provided to input new error compensation values as required.

3.4.16.14 Additional features. Unless otherwise specified (see 6.2.1), the following additional numerical control features shall be provided:

a. Incremental and continuous jog for each axis.
b. Leading zero suppression or decimal point programming.
c. Inches per minute (ipm)/millimeters per minute (mmpm) programming in all modes of operation.
d. Data edit keylock.

3.4.16.15 Operator control panel. Unless otherwise specified (see 6.2.1), the control unit shall have a control panel which provides at least the following control functions: Emergency Stop; Cycle Start; Optional

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Stop; Mode Selection; Feed Hold; Feedrate Override; Spindle Speed Override; Jog; Jog Direction; and, if necessary, EOB Stop.

3.4.16.15.1 Pendant or stanchion control. When specified (see 6.2.1), a pendant or stanchion control shall be furnished in lieu of an operator control panel. The pendant control shall be connected to the machine to enable the operator to control machine functions while in close observation position. Unless otherwise specified (see 6.2.1), the pendant control shall be capable of the same functions as stated for the operator control panel in paragraph 3.4.16.15.

3.4.16.16 Reference zero. The control unit shall have a reference zero to move each axis to a reference limit switch to establish synchronization and reset all registers to zero.

3.4.16.17 Program tryout. The control shall have a program tryout feature to permit new part programs to be read through the control without machine motions.

3.4.16.18 On-line system diagnostics. The MCU shall be capable of performing a constant monitoring function of various activities and conditions of the control and machine. The diagnostic system shall communicate appropriate fault messages to the operator by means of the CRT or other display. Unless otherwise specified (see 6.2.1), the on-line diagnostic capability shall determine the following fault conditions and provide appropriate error messages on the digital display:

- Axis feedback loop fail
- Illegal program entry
- Axis travel limits exceeded
- Low hydraulic pressure
- Axis servo motor failure
- Control console overheat
- Axis servo drive on/off
- Lubrication fault
- Guard door open
- Worn wheel
- Program error
- Feed limit

3.4.16.19 Maintenance diagnostics system. Maintenance diagnostics software shall be furnished as non-volatile memory or as specified (see 6.2.1) to provide for diagnostic checking of the machine control unit. The diagnostic software shall test, exercise, and display failures.

3.4.16.20 Peripheral equipment interface. When specified (see 6.2.1), the control shall be equipped with an interface complying with RS-232.

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3.4.16.21 Axis jog. The control panel or pendant shall have manual controls for accomplishing both single and continuous, low and high speed jog movements of the slides for each controlled axis in both the plus and minus direction.

3.4.16.22 Auxiliary functions. Unless otherwise specified (see 6.2.1), the MCU shall respond to the following program commanded auxiliary functions: interpolation, fixed cycles, and dwell. Unless otherwise specified (see 6.2.1), miscellaneous functions shall include program stop, optional program stop, end of program, coolant on-off, end of program (rewind), and program re-start.

3.4.16.23 Canned cycles. Canned cycles shall be provided as specified (see 6.2.1).

3.4.17 Drive mechanisms. Means shall be provided to position each moving member in all axes of motion. The wheel cross slide (X axis) on type I machines shall be positioned manually, utilizing handwheels and precision leadscrews; the wheelhead table (Z axis) shall be positioned by power. Type II, class A machines shall position along both X and Z axes by power. For power positioning, the system shall require that the operator set the location and, by a single push button, lever, or selector switch, cause the machine to locate the spindle at a chosen position in the workpiece. The system shall be a permanent part of the machine, protected from dust, swarf, or other foreign matter. The automatic control system shall have the ability, through operator control, to rapid traverse and decelerate to the actual desired position. Type II, class B machines shall position along each coordinate axis by numerical control. Rectilinear movement of the wheel cross slide (X axis) and wheelhead table (Z axis) shall be accomplished by closed loop, solid state DC servo systems. The servo motors shall be directly coupled to hardened and precision ground ball screws with recirculating ball screw nuts. The leadscrews shall be precision ground and handlapped to provide alignment and positioning accuracy. The X and Z axis feeds and positions are to be monitored by a feedback system and transmitted to the numerical control. Means shall be provided on all machines for proper lubrication. All slides and screws shall be shielded from contaminants by covers or guards.

3.4.18 Hour meter. When specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of one tenth of an hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching the maximum accumulative hours, the meter readout shall automatically revert to zero and continue to total time.

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3.4.19 Digital readout system. When specified (see 6.2.1), a digital readout system shall be installed on the X and Z axes of travel. The digital readout system shall consist of a position feedback device and electronic display for each axis specified and all related components, mounting hardware and electrical equipment necessary to assure complete operation of the system. The system shall be capable of providing a readily visible digital display over the full working range of each machine axis on which installed. The system resolution shall be 0.000025 inch. The system accuracy shall be 0.0002 inch per foot and repeatability shall be 0.000075 inch.

3.4.20 Position feedback device. A position feedback device including all related components, hardware and electrical equipment shall be provided for each axis having a readout system. The device shall provide axis travel feedback over the entire range of axis travel for each numerically controlled axis (see 3.4.16). Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device during operation of the machine. Operation of the position feedback device shall permit the machine to perform within the tolerances and accuracies given in table II.

3.4.21 Electronic display console. On type II, class B machines, the electronic display console shall contain a display with individual digits not less than 0.25 inch high. The console shall be capable of displaying at least six digits simultaneously. The display console shall also provide keyboard entry, memory, absolute-incremental counting, data preset, power failure indicator, inch/metric conversion and absolute zero recall. The electronic display console shall be installed so as to be readily accessible and easily read from the operator's work position. The console shall be mounted to withstand shock and vibration generated by the machine. The connecting cable shall be protected and shall be installed so as not to interfere with the normal operation of the machine. The cable shall be securely fastened to the structure of the machine in accordance with the requirements of the contract.

3.4.22 Standard equipment. Unless otherwise specified (see 6.2.1), the following standard equipment shall be furnished with each machine:

a. One set of hand tools normally furnished by the manufacturer.

b. One adjustable worklight, complete with protective shield and on-off switch.

c. One four-jaw, independent chuck equipped with standard three-step reversible top jaws. The chuck and top jaws shall conform to the applicable requirements of ANSI B5.8 for medium duty, wrench operated chucks. The chuck shall be fitted to the spindle flange and shall be the size specified (see 6.2.1).

d. One face plate of the size specified (see 6.2.1) having not less than two equally spaced slots.

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e. One extra set of workhead drive belts.

f. One extra set of wheelhead belts, if applicable (see 3.4.5).

g. Six grinding wheels of the type, dimensional characteristics and composition as specified (see 6.2.1).

h. Sine bar attachment to provide accurate taper grinding.

3.4.23 Optional equipment. Optional equipment and accessories shall be furnished as specified (see 6.2.1).

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and characteristics of the machine shall be not less than those specified in table I for the machine size specified.

3.6 Performance. Each machine shall be capable of grinding workpieces in accordance with requirements herein, and specifically in accordance with the test requirements in 4.4.

3.7 Alignment tolerances. Each machine shall meet the alignment tolerances of 4.4.5.

3.8 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.9 Marking of instruction plates, indicating plates and nameplates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

3.9.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.9.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2.1), the nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be abbreviated, provided the shortened entry for each caption is clear as to its identity.

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TABLE I. Sizes and capacities.

	Type I					Type II - Style A				
	Size 1	Size 2	Size 3	Size 4	Size 5	Size 1	Size 2	Size 3	Size 4	Size 5
CAPACITY:										
Swing inside standard guard, inches	8	12	24	36	42	8	12	24	36	42
Length of ground hole, range, inches	1/10 to 3	3/8 to 8	3/8 to 10	3/8 to 12	3/8 to 13	1-/10 to 3	3/8 to 5	5/8 to 15	3/8 to 24	3/8 to 13
Diameter of ground hole, range, inches	1/8 to 3-1/2	1/8 to 9	1/8 to 20	1/8 to 20	1/8 to 15	1/8 to 3-1/2	1/8 to 5-1/2	1/8 to 12	1/8 to 14	1/8 to 20
Taper, included angle, degrees	60	60	60	60	30	60	60	60	60	60
Diameter of workhead spindle flange, inches	5	9-5/8	9-5/8	12	12	6	6	9-5/8	12	12
Hole through workhead spindle, inches	15/16	1-1/2	1-11/16	1-11/16	1-11/16	15/16	1-1/2	1-11/16	1-11/16	1-11/16
Workslide (U axis) travel, manual or power, inches	12	12	12	12	12	12	12	12	12	12
Wheelhead table longtudinal (Z axis) travel, inches	6	16	15	18	20	6	9	15	18	20
Wheelhead cross slide (X axis) travel inches	2	4	12	12	12	2	3-1/4	6	12	12
SPEEDS AND FEEDS:										
Wheelhead table or assembly speed range, (X and Z axes), rapid traverse, IPM	0 to 350	0 to 350	0 to 200	0 to 200	0 to 200	0 to 350	0 to 350	0 to 200	0 to 200	0 to 200
Wheelhead table or assembly speed range, (X and Z axes), grinding, IPM	0 to 240	0 to 240	0 to 200	0 to 200	0 to 200	0 to 240	0 to 240	0 to 200	0 to 200	0 to 200
Wheelhead spindle speed, maximum, RPM	40,000	40,000	40,000	25,000	10,000	40,000	40,000	40,000	25,000	10,000
Workhead spindle speed, range of adjustment, RPM	960 or 1200	130, 200, 260, 400	40, 60, 80, 120	40, 60, 80, 120	40, 60, 80, 120	260 or 1200	130, 200, 260, 400	40, 60, 80, 120	40, 60, 80, 120	40, 60, 80, 120
MOTORS:										
Workhead drive, HP	1	1	1	1	1	3/4	1	1	1	1
Wheelhead drive, HP	5	5	7-1/2	7-1/2	7-1/2	5	5	7-1/2	7-1/2	7-1/2
Axes drives (X and Z axes), HP	—	—	—	—	—	1-1/2	1-1/2	1-1/2	1-1/2	1-1/2
**Hydraulic pump, HP	1-1/2	2	2	2	2	1-1/2	2	2	2	2
**Coolant pump, HP	1/4	1/2	1/2	1/2	1/2	1/4	1/2	1/2	1/2	1/2

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TABLE I. Sizes and capacities continued.

	Type II - Style B				
	Size 1	Size 2	Size 3	Size 4	Size 5
CAPACITY:					
Swing inside standard guard, inches	6	12	18	24	40
Length of ground hole, range, inches	1/16 to 9	1/16 to 9	1/16 to 17	1/16 to 16	1 to 20
Diameter of ground hole, range, inches	1/16 to 6	1/16 to 5-1/2	1/16 to 20	1/16 to 18	3-1/4 to 20
Taper, included angle, degrees	6	6	60	60	60
Diameter of workhead spindle flange, inches	5-1/2	9-5/8	9	9	12
Hole through workhead spindle, inches	3/4	1-1/2	1-1/2	1-1/2	1-11/16
Workslide (U axis) travel, manual or power, inches	4	8	12	10	16
Wheelhead table longtudinal (Z axis) travel, inches	8	12	20	20	30
Wheelhead cross slide (X axis) travel inches	4	6	9	10	14
SPEEDS AND FEEDS:					
Wheelhead table or assembly speed range, (X and Z axes), rapid traverse, IPM	0 to 350	0 to 500	0 to 350	0 to 350	0 to 350
Wheelhead table or assembly speed range, (X and Z axes), grinding, IPM	0 to 240	0 to 240	0 to 240	0 to 240	0 to 200
Wheelhead spindle speed, maximum, RPM	40,000	40,000	40,000	40,000	10,000
Workhead spindle speed, range of adjustment, RPM	30 to 3000	0 to 1000	30 to 900	30 to 900	0 to 200
MOTORS:					
Workhead drive, HP	3/4	1-1/2	4-1/2	4-1/2	4-1/2
Wheelhead drive, HP	10	5	10	10	10
Axes drives (X and Z axes), HP	1-1/2	1-1/2	1-1/2	1-1/2	1-1/2
**Hydraulic pump, HP	1/3	2	1/3	2	2
**Coolant pump, HP	1/4	1/2	1/2	1/2	1/2

** When applicable.

NOTE: Machine sizes and capacities shall be not less than the stated requirements of the size shown in the above table. When the largest size and capacity machine shown in above table is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated maximum or greater.

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Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.10 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrications charts, and machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD Form 1623. All technical data furnished shall be written in the English language.

3.11 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actua, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.2).

b. Quality conformance inspection (see 4.2.3).

4.2.1 Inspection conditions. Unless otherwise herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated

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indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.2.2 First article inspection. When a first article inspection is required under 3.1, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.3, all tests in 4.4 and the inspection in 4.5. Failure of the item to pass the first article tests shall be cause for rejection.

4.2.3 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of at least the examination in 4.3, the tests in 4.4 and the inspection in 4.5. Failure of the item to pass the examination or tests shall be cause for rejection.

4.3 Examination. The machine and its accessories shall be examined for design, construction, materials, components, electrical equipment, and workmanship to determine compliance with all of the requirements of this specification as given in 3.1 through 3.5 and 3.8 through 3.11.

4.4 Tests. The machine shall be subjected to the following tests. All instruments, tapes, materials, and tools required to perform and evaluate these tests shall be furnished by the supplier. The measuring instruments shall have calibration traceable to the National Bureau of Standards.

4.4.1 Operational test. The supplier shall demonstrate the machine's operational capabilities by performing the following test. During the warm-up period, proper operation of all manual controls, motors, adjustment mechanisms and accessories shall be verified. After warm-up, the machine shall be operated at no load for not less than four hours. This operation shall include spindle speed and feed rate changes that include the highest, intermediate, and lowest settings of each range, rapid traverse of both slides, simultaneous movement of slides, and automatic feed cycles, as applicable. The numerical control system on type II, class B machines shall be further tested to verify proper operation of MDI, program edit, and system diagnostics. Should a malfunction occur, it shall be corrected and the operational test repeated until a full four hours of running time is completed without failure.

4.4.2 Performance tests. The following operations shall be performed on a workpiece of mild steel. The nominal dimensions of the test piece shall be determined by the supplier prior to testing. The dimensions chosen should be based upon the size and capacity of the machine being tested and the test criteria detailed in paragraphs 4.4.2.2 and 4.4.2.3. The supplier shall choose the grinding wheel suitable for the particular program function to be tested. At the option of the procuring agency, each machine shall be subjected to grinding tests in accordance with specific requirements desig-

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nated by the activity in lieu of or in addition to the tests described herein (see 6.2.1).

4.4.2.1 Wheel dressing and truing test. When wheel dressing and truing cycles are numerically controlled (see 3.4.4), they shall be tested before performing the grinding tests in 4.4.2.2 and 4.4.2.3. The methods used in performing this test shall be the manufacturer's option provided they accurately determine the ability of the components and control unit to properly dress and true the grinding wheel.

4.4.2.2 Performance test for manual/power machines. To demonstrate the grinding capability of type I and type II, class A machines, a bore shall be ground not less than 2/3 the length of the capacity of the machine being tested as required under 1.2. The diameter of the bore shall be compatible with the size of the machine, the wheelhead, and tooling available.

4.4.2.3 Performance test for NC machines. For type II, class B machines, the following operations shall be performed on the workpiece. All operations are to be ground in one chucking.

a Grind a straight bore of the diameter determined by the manufacturer and not less than 1/2 the length of the capacity of the machine being tested as required under 1.2.

b. Grind a step bore with a diameter slightly larger than and concentric to the straight bore given in (a) above.

c. Face one end of the workpiece.

d. Grind the outside diameter of the end of the workpiece faced in (c) above.

4.4.2.4 Performance test requirements. Each machine tested must meet the following requirements, as applicable.

a. Straightness of all bores shall be within 0.00015 inch.

b. All bores shall be round within 0.000050 inch.

c. All bores shall be concentric within 0.00010 inch.

d. All internal and external diameters shall be ground to within $\pm$0.00020 inch of the nominal diameters.

e. Location of the ground face shall be within $\pm$0.00050 inch of the nominal location, flat within 0.00050 inch TIR and perpendicular to the centerline of the test part within 0.0002 inch per inch.

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f. Surface finish of all bores shall not be rougher than 10 micro-inch AA for machines sizes 1 through 3, and 18 micro-inch AA For machines sizes 4 and 5.

4.4.3 Positioning testing. For type II, class B machines, starting at positions other than the extremes of axis travel, two identical movements of both controllable linear axes shall be programmed and executed in each direction. Programmed span shall be not less than 80 percent of the maximum possible span, and each digit of starting and end points shall be other than zero with respect to reference zero Absolute positioning shall not exceed ±0.0003"/FT (not to exceed 0.025 inch in overall travel) and positioning repeatability shall not exceed ±0.000030 inch. Leadscrew reversal error shall be not greater than 0.000050 inch. Movement of each axis shall be programmed at feed rates approximating 10, 20, and 100 percent of maximum linear feed rate. Stops shall be programmed appropriately for inspection of control positioning accuracy. Positioning tests may be part of the same program as used in 4.4.1.

4.4.4 Overshoot test. On type II machines, overshoot of each axis shall not exceed 0.0005 inch at any feedrate up to 20 percent of the maximum. Maximum overshoot shall not exceed 0.001 inch.

4.4.5 Alignment accuracy tests. Each machine and its components shall be tested to determine that it complies with the alignment accuracy requirements of table II. The instruments and methods used in performing the tests shall be at the supplier's optional provided they accurately determine the tolerances of the machine.

4.4.6 Test program verification. Unless otherwise specified (see 6.2.1), the grinding test program (tape or other input data as specified in 3.4.16.1), shall be run on the machine to verify its ability to receive and perform all program functions without error. The supplier shall devise test set-ups to verify each program function.

4.4.7 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control testing shall be tested for compliance with 3.8.

4.5 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging, packing and marking. Unless otherwise specified (see 6.2.1), packaging, packing, and marking shall be in accordance with ASTM-D-3951. When specified (see 6.2.1), level A packaging, level A or level B packing and special marking shall be accomplished in accordance with MIL-M-18058.

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TABLE II - Alignment accuracy tests.

Characteristic	Permissible Tolerance
For type I and type II, class A machines:	
Base guideways	
- straight in longitudinal direction	0.0002"/10"
- flat in longitudinal direction	0.0002"/10"
Wheelhead table assembly, rise and fall, in its longitudinal (Z-axis) movement	0.0002"/10"
Wheelhead cross slide table assembly, rise and fall, in its traverse (X-axis) movement	0.0002"/10"
Workhead spindle parallel with table travel in the horizontal plane	0.0005"/10"
Workhead spindle flange, axial runout	0.00015" T.I.R.
Workhead spindle, axial slip (measured at two points displaced by 180°)	0.0002"
Workhead cross slide flat and parallel with table travel	0.0005"/10"
Workhead cross slide travel square with table travel	0.0005"/10"
For type II, class B machines:	
Base guideways	
- straight in longitudinal direction	0.0001"/10"
- flat in longitudinal direction	0.0001"/10"
Wheelhead table assembly, rise and fall in its longitudinal (Z-axis) movement	0.0001"/10"
Wheelhead cross slide table assembly, rise and fall, in its traverse (X-axis) movement	0.0001"/10"
Workhead spindle parallel with table travel in the horizontal plane	0.0003"/10"
Workhead spindle flange, axial runout	0.00010" T.I.R.
Workhead spindle, axial slip (measured at two points displaced by 180°)	0.00010"
Workhead cross slide flat and parallel with table travel	0.00025"/10"
Workhead cross slide travel square with table travel	0.00025"/10"

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6. NOTES

6.1 Intended use. The grinding machines covered by this specification are intended for use in any shop where large volumes of machined work parts are required to be ground. They are capable of grinding internal straight bores and tapered bores.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number, and date of this specification.

b. Type, class and size required (see 1.2).

c. First article approval, if required (see 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Specify additional safety and health requirements, if different (see 3.2.5 and 6.4).

g. Painting, if different (see 3.3.5).

h. Threaded parts, if different (see 3.3.6).

i. Dimensions for extended bridge type machine, if required (see 3.4.1).

j. Workhead cross slide power or numerically controlled operation, if required (see 3.4.2).

k. Power operated chuck, if required (see 3.4.3).

l. Programmable automatic swivel angle positioning of the workhead, if required (see 3.4.3).

m. Workhead spindle flange, if different (see 3.4.3.1).

n. Speed indicating device, if required (see 3.4.3.2).

o. Grinding wheel dresser, if not required; automatic "lift" positioning of grinding wheel dresser, if required (see 3.4.4).

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p. Specify wheelhead(s) and quills or sleeves required (see 3.4.5).

q. Numerically controlled wheel change, if required (see 3.4.8).

r. Coolant system, if not required (see 3.4.11).

s. Hydraulic system, if required (see 3.4.12).

t. Electrical system, electrical supply, and voltage for initial wiring, if different (see 3.4.13).

u. Motor requirements, if different (see 3.4.13.1).

v. Type(s) of automatic sizing system, if required (see 3.4.15).

w. Type of part program storage media required (see 3.4.16).

x. Data display, if different (see 3.4.16.1).

y. Memory type, if different; part program storage capacity, if different (see 3.4.16.4).

z. Standby battery power, if different (see 3.4.16.5).

aa. Numerically controlled automatic cycle, if required (see 3.4.16.11).

bb. Additional NC features, if different (see 3.4.16.14).

cc. Operator control panel functions, if different (see 3.4.16.15).

dd. Pendant control panel, if required, and control functions, if different (see 3.4.16.15.1).

ee. Fault conditions required, if different (see 3.4.16.18).

ff. Maintenance diagnostics software required (see 3.4.16.19).

gg. RS-232 interface, if required (see 3.4.16.20).

hh. Auxiliary functions, if different; miscellaneous functions, if different (see 3.4.16.22).

ii. Specify canned cycles, if required (see 3.4.16.23).

jj. Hour meter, if required (see 3.4.18).

kk. Digital readout system, if required (see 3.4.19).

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ll. Standard equipment, if different (see 3.4.22).

mm. Specify chuck size required (see 3.4.22.c).

nn. Specify size and number of slots for face plate (see 3.4.22.d).

oo. Grinding wheel types, dimensions, and composition required (see 3.4.22.g).

pp. Specify optional equipment, if required (see 3.4.23).

qq. Specify size and capacity, if different (see 3.5).

rr. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.8).

ss. Lubrication chart, if different (see 3.9.1).

tt. Nameplate, if different (see 3.9.2).

uu. First article inspection, if different (see 4.2.2).

vv. Quality conformance inspection, if different (see 4.2.3).

ww. Performance tests, if different (see 4.4.2).

xx. Grinding test program, if not required (see 4.4.6).

yy. Preservation, packaging, and marking, if different (see 5.1).

6.2.2 Contract data requirements. Required technical data, such as operator's manuals, parts lists, wiring diagrams, foundation and anchor bolt plans and acceptance test reports, should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment)

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the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. The machine shall be equipped with all points-of-operation guarding normally furnished as standard on the manufacturer's commercial machines supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required as, in most cases, the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements, should be specified in detail under 6.2.1(f).

6.5 Cross-reference of classification changes. Machine classification changes from the previous issue of this specification are as follows:

MIL-G-80066	MIL-G-80180B
Size 1	Type I, Size 1
Size 2	Type I, Size 2
Size 3	Type I, Size 3
Size 4	Type I, Size 4
Size 5	Type I, Size 5
Size 6	-----
Size 7	-----

MIL-G-80180A	MIL-G-80180B
Size 1	Type II, Class A, Size 1
Size 2	Type II, Class A, Size 2
-----	Type II, Class B, Sizes 1-4 New

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Preparing Activity:
DLA - IP

Project Number:
3415-0153

Review Activities:
Army - AL, ER
Navy - SH
Air Force - 84
DLA - GS

User Activities:
Army - SM, AR
Navy - MC, AS, OS, YD
Air Force - None

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-80180B	Grinding Machines, Internal, Chucking, Horizonta Electric Motor Driven

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

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